Gabay — Shipping a Public RAG Product

One-page reflection · Graded Mini Project 14.3 · 2026-10-11 · gabay.vercel.app · github.com/jacobvillar/gabay

Overview

Gabay is a public, domain-specific RAG assistant that answers questions about **Philippine statutory worker benefits** — 13th-month pay, leave, SSS, PhilHealth, and Pag-IBIG — strictly from official government handbooks, with the source passages shown under every answer. It is built with **Next.js + the Vercel AI SDK** (streaming chat, retrieval exposed as a tool call), **Groq** (open-weight GPT-OSS 120B) for generation, and **Upstash Vector** (built-in embedding model) for retrieval, deployed publicly on Vercel.

Corpus and key decisions

Decision	Choice	Why
Corpus	DOLE Handbook, SSS Citizens Charter, PhilHealth Konsulta circular, Pag-IBIG briefer	Public-domain PH government works — legal to serve publicly; a domain where base LLMs are unreliable, so retrieval + citations add real value
Generation	Groq — open-weight GPT-OSS 120B	Free tier, no OpenAI key, supports tool calling, deploys on Vercel
Embeddings	Upstash built-in model	Removes the embedding API key entirely — one fewer secret
Chunking	Per-page → ~1000-char windows, 150 overlap	Per-page parsing preserves real **page numbers** for citations
Metadata	`document_title`, `agency`, `page`, `source_url`	Readable citations that disambiguate across four agencies

The trickiest decision

Making retrieval **trustworthy on high-stakes questions**. A wrong peso figure or rule is worse than no answer, so retrieval had to be a deliberate **tool call** (not always-prepended context), with a tool description tuned so the model reaches for it on benefit questions but skips small talk, and a system prompt kept strict: answer only from retrieved passages, and say plainly when the handbooks do not cover something. Balancing that faithfulness against still being genuinely helpful was the hardest part to get right.

What it does well

Grounded, cited answers on PH-specific rules. "How is 13th-month pay computed?" triggers the retrieval tool, answers from the DOLE handbook (1/12 of basic salary, with the correct exclusions), and renders the exact source pages — something a raw model gets subtly wrong with no citations.

What it does badly

Retrieval balance and chunking. Because the SSS charter dominates the index, some non-SSS queries can surface less-relevant SSS chunks; character-based chunking occasionally splits a rule mid-sentence. These are quality, not correctness, issues — but they are real.

What I'd do with another week

1. **Rebalance the corpus** — trim the SSS charter to its benefits sections or add per-agency retrieval filters.
2. **Hybrid retrieval + reranking** — combine dense search with keyword matching and a reranker.
3. **Eval harness** — score faithfulness and context-relevance on the demo-question set.
4. **Inline citation mapping** — true [1]/[2] markers tied to the sources panel.